

GOPIREDDY SAHITI

CONTACT

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Bengaluru, India

TECHNICAL SKILLS

Languages & Environment

- Python
- Java & C/C++
- SQL & MATLAB

AI & Data Science

- Machine Learning
- Data Science & Analysis
- OOPs Principles
- Problem Solving

Cloud & Infrastructure

- AWS (Fundamentals)

Tools & Libraries

- Git & GitHub
- NumPy & Pandas
- Matplotlib & Seaborn

PROFESSIONAL SKILLS

- Collaboration
- Communication Skills
- Proactive Learning

AI/ML Engineering Student

Aspiring Software & Machine Learning Engineer

PROFILE

AIML undergraduate with practical experience in Data Science, Data Analysis, and building Machine Learning models from scratch in Python. Strong grounding in Object-Oriented Programming (OOPs), MATLAB, AWS cloud fundamentals, and database querying in SQL. Demonstrates excellent problem solving, proactive learning, and collaboration skills across technically rigorous software engineering and AI projects.

PROJECTS

Slot-Machine – Lightweight Interactive Python Game

- Developed a modular, object-oriented text-based slot machine simulation game in Python featuring dynamic betting logic and user balance tracking.
- Implemented core game mechanics including randomized multi-line symbol spinning, payline matching algorithms, and dynamic payout multiplier calculations.
- Designed robust user input validation and budget guardrails to ensure smooth terminal execution and problem-solving error handling.

DataVerse – Flight Delay Analytics & Predictive Modeling

- Executed end-to-end Data Science and Data Analysis pipelines on 5.8M U.S. domestic flight records to identify primary delay drivers and evaluate carrier reliability.
- Engineered exploratory data analysis (EDA) workflows and visual dashboards to evaluate carrier punctuality, airport congestion, and seasonal travel trends.
- Built and fine-tuned a machine learning arrival delay prediction model achieving an R^2 score of 0.934, predicting flight delays within an average margin of 9.8 minutes.

Linear Regression – Custom Implementation from Scratch

- Engineered Simple and Multiple Linear Regression algorithms completely from scratch using NumPy and Pandas applying OOPs design principles.
- Implemented custom vectorized gradient descent optimization routines, feature scaling/normalization pipelines, and cost function tracking.
- Evaluated model performance using regression metrics (MSE, MAE, R^2) and created Matplotlib visualization scripts to plot convergence loss curves.

EDUCATION

New Horizon College of Engg.

B.E. in AI & Machine Learning

CGPA: 9.10 • 4th Semester

Vaibhav Junior College

MPC — 12th Grade • 94%

St. Michael's School

CBSE — 10th Grade • 83.8%

CERTIFICATIONS

Dataverse — Data Analytics Event

Certificate of Achievement

NPTEL: Introduction to ML

Online Certification Course

Spoken Tutorial — Git Training

IIT Bombay Spoken Tutorial

INTERESTS

- Data Science & Visualisation
- Machine Learning from Scratch
- Algorithmic Problem Solving
- Software Development

Logistic Regression – Binary Classification from Scratch

- Developed a modular binary classification Machine Learning model using Logistic Regression implemented from scratch with NumPy and Pandas.
- Trained and evaluated the model on the Breast Cancer Wisconsin dataset using z-score feature scaling, sigmoid activation, and vectorized gradient descent.
- Computed key evaluation metrics (Accuracy, Precision, Recall, F1-Score) and plotted decision boundaries and training loss curves using Matplotlib.

MediAssist AI – Safety-Aware Healthcare Triage Framework

- Designed a hybrid AI healthcare triage framework combining deterministic safety rules with Machine Learning.
- Implemented pre-inference emergency escalation for critical symptom detection using structured data analysis.
- Developed a post-inference medical consistency validation layer to improve prediction reliability.
- Designed a defence-in-depth architecture for robust decision-making on sparse medical data.
- Performed model evaluation and error analysis using standard classification metrics.

CORE COMPETENCIES

- Strong foundation in Data Science, Machine Learning algorithms, OOPs principles, and Problem Solving across Python, Java, C/C++, and MATLAB.
- Hands-on experience conducting Data Analysis and building ML models from scratch with NumPy/Pandas on multi-million row datasets (5.8M records).
- Working knowledge of relational databases, schema design, query writing using SQL, and AWS cloud fundamentals.
- Strong Communication Skills, active Collaboration mindset, and proactive version control workflows using Git.

WHAT I BRING

- Showcased strong initiative across 5 end-to-end projects spanning Data Science, healthcare triage AI, flight analytics, and scratch ML algorithms.
- Comfort moving seamlessly between low-level logic (C/C++ data structures, OOPs) and high-level data science pipelines (Python/NumPy/Pandas/MATLAB).
- Proactive Learning habit reflected in top academic standing (9.10 CGPA) and multi-domain certifications in ML and Git.
- Strong Collaboration and Communication Skills paired with an eye for clean, user-friendly data visualizations.